Student Number _____

Student Name _____

The Chinese University of Hong Kong, Shenzhen

FINAL EXAMINATION

Term 2, Spring, 2021-2022

CSC3002 Introduction to Computer Science:

Programming Paradigms

Examination Duration: 120 minutes.

This examination has 32 questions divided into 4 sections.

Exam Conditions:

This is a FORMAL Examination.

This is a RESTRICTED OPEN BOOK Examination.

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of double-sided A4 paper notes are permitted. NO OTHER MATERIALS ARE PERMITTED

Calculators are NOT ALLOWED in this examination.

2B pencils and erasers should be brought by the students and used to mark the answer sheet.

Materials To Be Supplied To Students:

This question and answer book, the answer sheet, and scratch paper.

Materials To Be Collected From Students:

EVERYTHING MUST BE RETURNED, including this question and answer book, the answer sheet, and scratch paper. Please answer the questions in the first 3 sections by marking the answer sheet, and those in the last section by writing down your answers in the specified blanks in this question and answer book.

Section 1. True or false questions. (10 × 1% = 10%)
Pick "a" for "True" or "b" for "False" for each of the following statements. Please mark your answers (only "a" or "b") on the answer sheet.

1. In C++, a **variable** is a named address for storing a type of value, and a **pointer** is a special type of variable that holds the address of another variable.

   a. True b. False

2. In C++, a **function** is a named section of code that performs a specific operation, and a function must have a parameter list as the input and return a value as the output.

   a. True b. False

3. The following two statements will generate two strings with the same content:

```
std::string str1 = "hello,world";
std::string str2 = "hello" + "," + std::string("world");
```

   a. True b. False

4. In C++, streams are complicated objects that cannot be copied. If you want to pass a stream object into a function you must use **call by reference**, and if you want to return a stream object from a function, you must use **return by reference**.

   a. True b. False

5. In an array, the size of the array specified in the declaration is called the **allocated size**. The number of elements actively in use is called the **effective size**. The C++ compiler makes sure that the effective size can never exceed the allocated size.

   a. True b. False

6. The following statements declare two pointers **`p1`** and **`p3`**, and two integers **`p2`** and **`p4`**:

```
int*  p1, p2;
int  *p3, p4;
```

   a. True b. False

7. For a variable **`x`**, the expression **`*&x`** is essentially equivalent to **`x`**; and for a pointer **`p`**, the expression **`&*p`** is equivalent to **`p`**.

   a. True b. False

8. A **hash function** should distribute keys as uniformly as possible across the integer range to avoid **collisions**, and the **`put`** and **`get`** operations can only achieve $O(1)$ performance without any collision.

   a. True b. False

9. If **`c`** is a nonempty collection in C++ STL, calling **`c.begin()`** and **`c.end()`** will return **iterators** pointing at the first and last element of that collection, respectively.

   a. True b. False

10. C++ is a **multi-paradigm** programming language. Besides the **procedural** programming paradigm originally supported by C, C++ also supports the **object-oriented** paradigm. However, C++ only supports other paradigms such as the **functional** paradigm through external libraries.

    a. True b. False

Section 2. Single choice questions. (10 × 2% = 20%)
Pick the correct option in each of the following questions. Note that only ONE option is correct. Please mark your answers on the answer sheet.

11. Calculate the result of the following expressions based on the C++ rules:

```
9.0*(8-7) / 6 + 5*4 % 3*(2+1)
```

    a. **3**
    b. **3.5**
    c. **7**
    d. **7.5**

12. A local variable in a function is usually destroyed when the function returns. If you want to retain its value until the next time the function is called, which keyword should be used before the declaration of the variable:

    a. const
    b. extern
    c. static
    d. protected

13. Suppose that you are using the **Merge sort** algorithm to sort a vector of 1,000 integers and find that it takes T milliseconds to complete the operation. How many milliseconds would you expect the running time to be if you use the same algorithm to sort a vector of 1,000,000 integers on the same machine? (Assuming there is enough memory and $\log_2 1000 \approx 10$.)

    a. 2,000 T
    b. 10,000 T
    c. 20,000 T
    d. 1,000,000 T

14. If we have built a 40-bit computer on which the memory address will be represented in one 40-bit long **machine word**, and the **smallest addressable unit** is a sequence of 8 consecutive bits called a **byte**, how much memory can this computer manage, theoretically? (Assume 1KB = 2^{10} bytes, 1MB = 2^{10} KB, 1GB = 2^{10} MB.)

    a. 40 GB
    b. 256 GB
    c. 320 GB
    d. 1024 GB

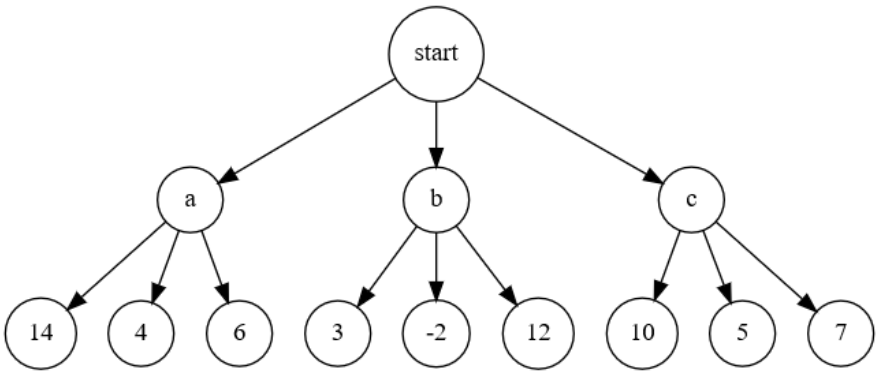
15. Assuming the following C++ variable declaration statement

```
int i = 2022, *pi = &i, **ppi = &pi;
```

leads to the addresses of **`i`**, **`pi`**, and **`ppi`** being **`0x5190`**, **`0x5198`**, and **`0x51A0`**, respectively, what are the values of **`**ppi`** and **`*ppi`**?

a. **`0x51A0`** and **`0x5198`**
b. **`0x5198`** and **`0x5190`**
c. **`0x5190`** and **`0x5198`**
d. **`2022`** and **`0x5190`**

16. In a two-player game, suppose you are in a position in which the analysis for the next two moves shows the following rated outcomes (larger values represent better outcomes for you) from your original player's point-of-view:



If you adopt the **minimax** strategy in your program, what is the best move and the rating of that move from your perspective?

a. **`a, 14`**
b. **`b, 12`**
c. **`c, 5`**
d. **`b, -2`**

17. Computers typically represent each pixel in an image as a 32-bit integer or 8 hexadecimal digits (starting with **`0x`**), in which the 4 bytes are interpreted as the transparency (α) and color (Red, Green, Blue) components. The following bitwise operations extract one of the components from the hexadecimal representation of the αRGB color stored in an integer variable **`pixel`**. Which component is extracted and what is the value of that component in **decimal** format?

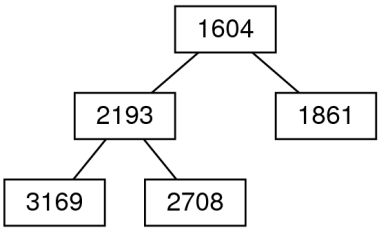
```
int pixel = 0x05192022;
int color = (pixel >> 8) & 0xFF;
```

a. α, 0
b. α, 256
c. Green, 20
d. Green, 32

18. If we build a binary search tree by inserting the nodes in increasing order (i.e., the smaller the key is, the earlier the node is inserted), and do not balance it, the result will be an extremely unbalanced tree growing linearly. If we then traverse this tree and **`cout`** the key, which one of the following traversal approaches will have different output from the others?

a. Preorder
b. Inorder
c. Postorder
d. Level-order

19. Suppose that you are modelling a **priority queue** with a **partially ordered tree** that contains the following data (assuming smaller values have higher priorities):



What is the proper content in the corresponding **heap** after deleting a node and then inserting "2022"?

a. 1604, 2022, 1861, 3169, 2193
b. 1861, 2022, 2193, 2708, 3169
c. 1861, 2022, 2708, 3169, 2193
d. 1861, 2193, 2022, 3169, 2708

20. What does the following declaration do?

```
double (*f)(double)
```

a. Declares **`f`** as a function taking a double and returning a pointer to a double.
b. Declares **`f`** as a function object or functor taking a double and returning a double.
c. Declares **`f`** as a pointer to a function taking a double and returning a double.
d. Declares **`*f`** as a function taking a double and returning a double.

Section 3. Multiple choice questions. (10 × 4% = 40%)
Pick the correct option(s) in each of the following questions. Note that there may be ONE to FOUR correct options for each question. Incomplete answers will get partial scores (2%). Please mark your answers on the answer sheet.

21. What are the **distinguishing** features of **object-oriented** programming languages like C++ (i.e., what makes object-oriented programming different from other programming paradigms)?

a. Abstraction
b. Encapsulation
c. Inheritance
d. Polymorphism

22. In C++, functions can be **overloaded**, which means that you can define different functions with the same name as long as they have different **signatures**. The signature of a function includes the following:

a. The number of the parameters
b. The names of the parameters
c. The types of the parameters
d. The type of the return value

23. Which of the following collection classes do not support the use of the range-based **`for`** loop?

a. **`vector`**
b. **`stack`**

c. **`queue`**
d. **`map`**

24. In the definition of a class, the data members can be declared as:

    a. public
    b. protected
    c. friend
    d. private

25. Normally in C++ you should not use pointers as much as in the C language, in which pointers are important for the following reasons:

    a. Pointers allow you to specify the memory addresses for dynamically allocated variables and objects in the heap.
    b. Pointers make it possible to reserve new memory during program execution.
    c. Pointers allow you to refer to a large data structure in a compact way.
    d. Pointers can be used to record relationships among data items, e.g., linked list.

26. In C++, any expression that refers to an internal memory location capable of storing data is called an **lvalue**, which can appear on the left side of an assignment statement. Which of the following terms in a C++ program **cannot** be lvalues?

    a. **`p`**
    b. **`(p+1)`**
    c. **`&p`**
    d. **`*p`**

27. What **properties** must a problem have for **recursion** to make sense as a solution strategy?

    a. There must be a function that calls itself.
    b. There must be an inclusion-exclusion pattern in the problem.
    c. There must be simple cases or exit points for which the answer is easily determined, without using recursion.
    d. There must be a recursive decomposition that allows you to break any complex instance of the problem into simpler problems of the same form.

28. A **ring buffer** is an array whose end is linked back to the beginning. In a typical ring buffer based queue implementation as follows,

```
ValueType *array;
int capacity;
int head;
int tail;
```

    the **`head`** field holds the index of the next element to come out of the queue, and the **`tail`** field holds the index of the next free slot. If we use the equality of the **`head`** and **`tail`** fields to indicate an empty queue, which of the following cases may indicate that the queue is full?

    a. **`head == tail`**
    b. **`head - tail == 1`**
    c. **`tail - head == capacity`**
    d. **`tail - head == capacity - 1`**

29. To avoid the default shallow copy or slicing in C++, which of the following functions should be overloaded when designing new classes:

    a. The copy constructor
    b. The assignment operator (**=**)
    c. The insertion operator (**<<**)
    d. The selection operator (**[]**)

30. After running the following program, which of the numbers will be seen in the output?

```
#include <iostream>
using namespace std;
class A {
public:
    int a;
    A() { a = 1; }
    virtual void display() { cout << a << endl; }
};
class B: public A {
public:
    int b;
    B() { b = 2; }
    void display() { cout << a << b << endl; }
};
class C: public B {
public:
    int c;
    C() { a = 2; c = 3; }
    void display() { cout << a << b << c << endl; }
};
int main() {
    C oC;
    B* pb = &oC;
    pb->display();
    return 0;
}
```

    a. **1**
    b. **2**
    c. **3**
    d. None, compile error.

Section 4. Fill-in-the-Blank Questions. (12%+18% = 30%)
Answer the following questions. Please write your answers on THIS EXAM PAPER in the blank space after the questions.

31. What is the formatted output of the following program? Write your answer in the box, and make sure the format is correct. (12%)

```
#include <iostream>
using namespace std;
int main(void)
```

```
{
    int arr[] = {0, 5, 1, 9};
    int* p = arr;
    int a = *++p;
    int b = ++*p;
    int c = *p++;
    int d = (*p)++;
    cout << a << b << c << d << endl;
    cout << arr[0] << arr[1] << arr[2] << arr[3] << endl;
    return 0;
}
```

32. Assume that **doubleArray** is declared as

```
double doubleArray[] = {1, 3, 5, 7, 9, 11, 13, 15, 17, 19};
```

and variables of type double take up **8 bytes** on the computer system you are using. If the base address of the array **doubleArray** happens to be **0x5190**, what are the values of the following 12 expressions? Write your answer in the boxes below, **all in 4-character hexadecimal format**, e.g., **10** should be **000A**. (12 x 1.5% = 18%)

doubleArray	= 0x	5190
&doubleArray	= 0x	
*doubleArray	= 0x	
doubleArray+1	= 0x	
&doubleArray+1	= 0x	
*doubleArray+1	= 0x	
doubleArray[2]	= 0x	
&doubleArray[2]	= 0x	
doubleArray+8	= 0x	
*doubleArray+8	= 0x	
*(doubleArray+8)	= 0x	
doubleArray[9]	= 0x	
&doubleArray[9]	= 0x	

END OF EXAMINATION